Lecture 10

Momentum maps and reduction

Objectives. *Separating the central force in Lecture 9 produced the constants L_3 and $|\mathbf{L}|^2$, and they came from the rotations. This lecture turns a whole group of canonical symmetries into conserved quantities. The* momentum map *collects observables whose flows generate the group action. For lifted transformations of positions, its components are Noether charges $p \cdot \xi$. For rotations it is $\mathbf{L} = q \times p$. Conserved quantities then* reduce *the motion. A cyclic variable removes a degree of freedom, and the central force reduces to one degree of freedom. Kepler's problem has an additional conserved vector that is not the charge of any point symmetry. It explains why bounded orbits with nonzero angular momentum close.*

By the end of this lecture you will be able to:

- Compute the momentum map of a cotangent-lifted group action.
- Reduce a Hamiltonian system using a cyclic coordinate.
- Derive the effective radial dynamics of a central force.
- Classify bounded Kepler orbits with nonzero angular momentum using the Laplace–Runge–Lenz vector.

10.1 Matrix groups and their generators

A change of notation. From this lecture to the end of the course the letter J is the *momentum map*. The matrix $\left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$ of Lectures 4 and 6 is written $\mathbb{J}$, so that

$$\omega(u,v) = u^{\mathsf{T}}\mathbb{J}v, \qquad X_f = \mathbb{J}\nabla f = \Big(\frac{\partial f}{\partial p}, -\frac{\partial f}{\partial q}\Big), \qquad X_f h = \{h, f\}\,. \tag{10.1}$$

Every phase space in this lecture is an open subset of $\mathbb{R}^{2n}$ with the canonical bracket, except that a coordinate may be an angle.

The groups of symmetries we need are groups of matrices, and everything we need about them is built from the exponential.

Definition 10.1 (Matrix group and its Lie algebra)**.** A *matrix group* is a subgroup G of the group $\mathrm{GL}(N,\mathbb{R})$ of invertible $N \times N$ matrices that is closed in it: a limit in $\mathrm{GL}(N,\mathbb{R})$ of matrices of G lies in G. Its *Lie algebra* is

$$\mathfrak{g} = \{\xi \in \mathbb{R}^{N\times N} \,:\, e^{s\xi} \in G \text{ for all } s \in \mathbb{R}\},$$

the set of ξ whose *one-parameter subgroup* $s \mapsto e^{s\xi}$ lies in G.

The Lie algebra $\mathfrak{g}$ is a real vector space and is closed under the commutator $[\xi,\eta] = \xi\eta - \eta\xi$. Both facts are standard for matrix groups; we *verify them directly* for the two groups the lecture uses.

- $\mathbb{R}^d$, acting on itself by translations. As matrices, $a \mapsto \left(\begin{smallmatrix} I & a \\ 0 & 1 \end{smallmatrix}\right)$; $\mathfrak{g} = \mathbb{R}^d$ with $[\xi, \eta] = 0$.
- $\mathrm{SO}(3) = \{R : R^{\mathsf{T}} R = I,\ \det R = 1\}$. Differentiating $e^{s\xi^{\mathsf{T}}} e^{s\xi} = I$ at $s = 0$ gives $\xi^{\mathsf{T}} + \xi = 0$; conversely, if $\xi^{\mathsf{T}} = -\xi$ then $\frac{\mathrm{d}}{\mathrm{d}s}\left(e^{s\xi}\right)^{\mathsf{T}} e^{s\xi} = 0$ for all s, so $e^{s\xi} \in \mathrm{O}(3)$, and $\det e^{s\xi} = e^{s \operatorname{tr} \xi} = 1$. Thus $\mathfrak{so}(3)$ is the space of 3×3 skew matrices, a vector space of dimension 3, and it is closed under the commutator because $[\xi, \eta]^{\mathsf{T}} = \eta^{\mathsf{T}} \xi^{\mathsf{T}} - \xi^{\mathsf{T}} \eta^{\mathsf{T}} = \eta\xi - \xi\eta = -[\xi, \eta]$ for skew ξ, η.

Proposition 10.2 ($\mathfrak{so}(3)$ is $(\mathbb{R}^3, \times)$)**.** *The map $\xi \mapsto \hat{\xi}$ with $\hat{\xi} v = \xi \times v$, that is*

$$\hat{\xi} = \begin{pmatrix} 0 & -\xi_3 & \xi_2 \\ \xi_3 & 0 & -\xi_1 \\ -\xi_2 & \xi_1 & 0 \end{pmatrix},$$

is a linear bijection of $\mathbb{R}^3$ onto $\mathfrak{so}(3)$, and

$$\left[\hat{\xi}, \hat{\eta}\right] = \widehat{\xi \times \eta}. \tag{10.2}$$

*For $|n| = 1$, $e^{s\hat{n}}$ is the rotation about n through the angle s, counter-clockwise seen from the tip of n; and every $R \in \mathrm{SO}(3)$ is of this form (*Euler's axis theorem*).*

Proof. The displayed matrix is skew, the map is linear, and it is onto because a skew 3×3 matrix has exactly three free entries. For (10.2), apply both sides to v: $\xi \times (\eta \times v) - \eta \times (\xi \times v) = (\xi \times \eta) \times v$ by the Jacobi identity of the cross product. Next, let $|n| = 1$. Since $\hat{n} v = n \times v$ and $n \times (n \times w) = -w$ for $w \perp n$, we have $\hat{n}^3 = -\hat{n}$. So in the exponential series the odd powers of $s\hat{n}$ sum to $\sin s\, \hat{n}$ and the even powers to $I + (1 - \cos s)\hat{n}^2$. This is Rodrigues' formula

$$e^{s\hat{n}} = I + \sin s\, \hat{n} + (1 - \cos s)\, \hat{n}^2.$$

For a unit $u \perp n$ it gives

$$e^{s\hat{n}} n = n, \qquad e^{s\hat{n}} u = \cos s\, u + \sin s\, n \times u, \qquad e^{s\hat{n}}(n \times u) = \cos s\, n \times u - \sin s\, u.$$

This is the rotation about n through s.

Finally let $R \in \mathrm{SO}(3)$. Then $\det(R - I) = \det\big(R(I - R^{\mathsf{T}})\big) = \det(I - R) = -\det(R - I)$, so $\det(R - I) = 0$, and $Rn = n$ for some unit n. Being orthogonal, R maps the plane $n^{\perp}$ to itself; in an orthonormal basis $(n, u, n \times u)$ its matrix is therefore $1 \oplus R_2$ with R_2 orthogonal and $\det R_2 = \det R = 1$, so R_2 is the rotation of the plane through some angle s. By the formulas above, $e^{s\hat{n}}$ has the same matrix. ∎

Definition 10.3 (Action; symplectic action; generator)**.** Let $\mathcal{M} \subseteq \mathbb{R}^{2n}$ be open. An *action* of a matrix group G on $\mathcal{M}$ is a map $\Phi\colon G \times \mathcal{M} \to \mathcal{M}$, smooth in both arguments, with

$$\Phi_e = \mathrm{id}, \qquad \Phi_g \circ \Phi_h = \Phi_{gh} \quad (g, h \in G).$$

It is *symplectic* if every Φ_g is a canonical map (Definition 8.1). The *infinitesimal generator* of $\xi \in \mathfrak{g}$ is the vector field

$$\xi_{\mathcal{M}}(x) = \left.\frac{\mathrm{d}}{\mathrm{d}s}\right|_{s=0} \Phi_{e^{s\xi}}(x). \tag{10.3}$$

Each Φ_g is a bijection, with inverse $\Phi_{g^{-1}}$. The map $\xi \mapsto \xi_{\mathcal{M}}(x)$ is linear. For the two groups of this lecture this is visible in the formulas, and in general it is a standard fact about matrix groups.

For example, rotations act on $\mathbb{R}^6$ by $\Phi_R(q,p) = (Rq, Rp)$. By Proposition 10.2, the generator of $\hat{\xi}$ is $\xi_{\mathcal{M}}(q,p) = (\xi \times q, \xi \times p)$.

Lemma 10.4 (The flow of an infinitesimal generator · *proved modulo T1 (cited)*)**.** *The flow of $\xi_{\mathcal{M}}$ is $s \mapsto \Phi_{e^{s\xi}}$, and it is complete.*

Proof. Let $\gamma(s) = \Phi_{e^{s\xi}}(x)$. By the group law, $\gamma(s+\sigma) = \Phi_{e^{\sigma\xi}}(\gamma(s))$, so $\gamma'(s) = \frac{\mathrm{d}}{\mathrm{d}\sigma}|_0 \Phi_{e^{\sigma\xi}}(\gamma(s)) = \xi_{\mathcal{M}}(\gamma(s))$, and $\gamma(0) = x$. It is defined for every s because $e^{s\xi} \in G$ for every s. By uniqueness (T1, Theorem 1.3) it is *the* integral curve. ∎

10.2 The momentum map and cotangent lifts

Definition 10.5 (Momentum map)**.** Let G act symplectically on $\mathcal{M}$, with $\mathfrak{g}$ of dimension d, and fix a basis $\xi^1, \dots, \xi^d$ of $\mathfrak{g}$. A *momentum map* for the action is a smooth map $J = (J_1, \dots, J_d) \colon \mathcal{M} \to \mathbb{R}^d$ such that for every $\xi = \sum_a c_a \xi^a \in \mathfrak{g}$ the observable

$$J_\xi := \sum_a c_a J_a \quad \text{satisfies} \quad X_{J_\xi} = \xi_{\mathcal{M}}. \tag{10.4}$$

For rotations the observable will be $J_\xi = \xi \cdot \mathbf{L}$, as Lecture 6 suggests and Example 10.8 verifies. So J is a vector of d observables, one for each basis direction of the algebra, and *each direction of the group is the flow of its own observable*: $\varphi_s^{J_\xi} = \Phi_{e^{s\xi}}$ by Lemma 10.4. Since $\xi \mapsto \xi_{\mathcal{M}}$ and $\xi \mapsto J_\xi$ are both linear, it is enough to check (10.4) on the basis. On a connected $\mathcal{M}$, the momentum map is unique up to a constant vector. Indeed, if J and J' are two momentum maps, then $J_\xi - J'_\xi$ generates the zero field, so it is constant by Corollary 6.8.

Cotangent lifts: every symmetry of positions has one

The main source of momentum maps is Lecture 6's lift. Let $U \subseteq \mathbb{R}^n$ be open and let G act on U by diffeomorphisms h^g, with generators $\xi_U \in C^\infty(U, \mathbb{R}^n)$. Proposition 6.10 lifts a one-parameter family h^s of point transformations to $\mathcal{M} = U \times \mathbb{R}^n$; the whole group lifts in the same way,

$$\Phi_g(q,p) = \big(h^g(q),\ Dh^g(q)^{-\mathsf{T}} p\big). \tag{10.5}$$

This is a group action (the chain rule applied to $h^{gg'} = h^g \circ h^{g'}$), and each Φ_g is canonical, being a lifted point transformation, as shown in Example 8.11.

Theorem 10.6 (Momentum maps of lifted actions)**.** *The action (10.5) has the momentum map*

$$J_\xi(q,p) = p \cdot \xi_U(q). \tag{10.6}$$

Proof. By (10.1),

$$X_{J_\xi} = \Big(\frac{\partial J_\xi}{\partial p},\ -\frac{\partial J_\xi}{\partial q}\Big) = \big(\xi_U(q),\ -D\xi_U(q)^{\mathsf{T}} p\big).$$

Now compute $\xi_{\mathcal{M}}$ from (10.5). Write $h^s = h^{e^{s\xi}}$; as in Lemma 10.4 the group law makes $s \mapsto h^s(q)$ the integral curve of ξ_U through q, defined for all s, so $h^0 = \mathrm{id}$ and $Dh^s(q) = I + s\,D\xi_U(q) + O(s^2)$, whence $Dh^s(q)^{-\mathsf{T}} = I - s\,D\xi_U(q)^{\mathsf{T}} + O(s^2)$. Differentiating (10.5) at $s = 0$,

$$\xi_{\mathcal{M}}(q,p) = \big(\xi_U(q),\ -D\xi_U(q)^{\mathsf{T}} p\big) = X_{J_\xi}(q,p).$$

Finally $\xi \mapsto p \cdot \xi_U(q)$ is linear, as Definition 10.5 requires. ∎

The formula gives Noether's charge from Lecture 3: $p \cdot \xi$ is what Theorem 3.14 produces from a symmetry of $\mathcal{L}$ and shows to be conserved. Proposition 6.10 already showed that it generates the lifted symmetry. What Theorem 10.6 adds is that the whole group is covered at once.

Example 10.7 (Linear momentum)**.** For translations of $U = \mathbb{R}^n$, the action is $h^a(q) = q + a$. Thus $\xi_U(q) = \xi$ is constant and $Dh^a = I$, so the lifted action is $(q, p) \mapsto (q + a, p)$. By (10.6), $J_\xi = p \cdot \xi$, so $J = p$.

Example 10.8 (Angular momentum)**.** For rotations of $U = \mathbb{R}^3$, the action is $h^R(q) = Rq$. Thus $\xi_U(q) = \xi \times q$ by Proposition 10.2, and $Dh^R = R$ is orthogonal, so $Dh^{R-\mathsf{T}} = R$: the lifted action is $(q, p) \mapsto (Rq, Rp)$, with generator $\xi_{\mathcal{M}}(q, p) = (\xi \times q,\ \xi \times p)$. Then

$$J_\xi = p \cdot (\xi \times q) = \xi \cdot (q \times p), \qquad \text{so} \qquad J = \mathbf{L} = q \times p. \tag{10.7}$$

So the flow of $J_n = n \cdot \mathbf{L}$, $|n| = 1$, is the rotation $e^{s\hat{n}}$ applied to q and p together, counterclockwise about n; Example 6.2 saw it for $n = e_3$. Figure 10.1 draws it.

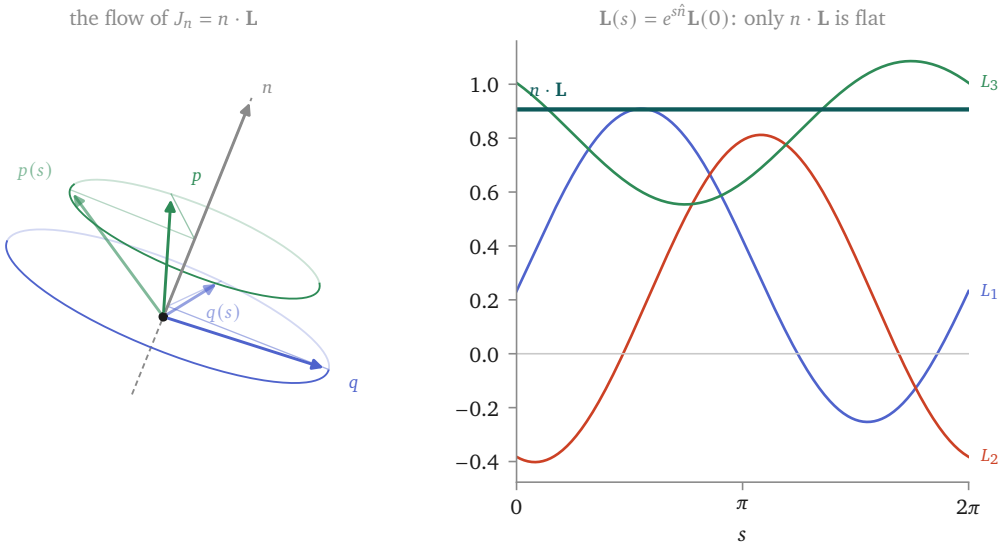


Figure 10.1: The flow of $J_n = n \cdot \mathbf{L}$, $|n| = 1$. Left: q and p are carried by *one* rotation about n, each round its own circle, so their lengths and the angle between them do not change. Right: $\mathbf{L}$ is a vector (Proposition 5.14), so it turns with them, $\mathbf{L}(s) = e^{s\hat{n}}\mathbf{L}(0)$. The component $n \cdot \mathbf{L}$ — the generator itself — is constant; the rest of $\mathbf{L}$ turns unless $\mathbf{L}$ is parallel to n.

10.3 Noether for groups

Theorem 10.9 (Noether for groups · *proved; (ii)⇒(i) uses T1 (cited), through Theorem 6.9*)**.** *Let G act symplectically on $\mathcal{M}$ with momentum map J, let $H \in C^\infty(\mathcal{M})$, and suppose every element of G is an exponential e^ξ, $\xi \in \mathfrak{g}$. The following are equivalent:*

(i) $\{J_\xi, H\} = 0$ *for every* $\xi \in \mathfrak{g}$*;*

(ii) every component of J is conserved by the flow of H;

(iii) H is G-invariant: $H \circ \Phi_g = H$ for every $g \in G$.

In that case every motion of H stays inside one level set $J^{-1}(\mu)$.

Proof. (i)⇔(ii) is Theorem 6.9 applied to each component J_ξ. For a *fixed* ξ, the same theorem's (i)⇔(iii) says that $\{J_\xi, H\} = 0$ holds exactly when H is invariant under $\varphi_s^{J_\xi} = \Phi_{e^{s\xi}}$ (Lemma 10.4), for all s. By the hypothesis on G, every Φ_g is such a map, with $s = 1$, so invariance under all of them is invariance under all of G; conversely (iii) contains each of them. Finally, J is constant along each motion, so the motion stays in the level set through its initial state. ∎

The hypothesis holds for both groups of the lecture. The translation by b is $\exp\left(\begin{smallmatrix} 0 & b \\ 0 & 0 \end{smallmatrix}\right) = \left(\begin{smallmatrix} I & b \\ 0 & 1 \end{smallmatrix}\right)$, the matrix being square-zero. Every rotation is $e^{s\hat{n}}$ by Euler's axis theorem (Proposition 10.2). So for the rotations of Example 10.8: *H is invariant under every rotation exactly when all three components of $\mathbf{L}$ are conserved.* A central force, $H = |p|^2/2m + V(|q|)$, is invariant, because $|Rq| = |q|$ and $|Rp| = |p|$.

10.4 Cyclic variables

The simplest use of a conserved quantity is a momentum whose coordinate is missing from H. Write $x = (q_1, q', p_1, p')$ with $q' = (q_2, \dots, q_n)$, $p' = (p_2, \dots, p_n)$, and let $\mathcal{M} = S \times \mathcal{M}'$, where q_1 ranges over all of $S = \mathbb{R}$ (or over a circle $S = S^1$, if q_1 is an angle) and (q', p_1, p') over an open set $\mathcal{M}'$. The product form matters: it lets the translations of q_1 act on all of $\mathcal{M}$.

Theorem 10.10 (Reduction by a cyclic variable)**.** *Suppose $H = H(q', p_1, p')$ does not depend on q_1. Along every motion:*

1. *p_1 is constant, say $p_1 = \ell$;*
2. *$(q'(t), p'(t))$ solves Hamilton's equations for the* reduced Hamiltonian *$H_\ell(q', p') = H(q', \ell, p')$, in $n-1$ degrees of freedom;*
3. *q_1 is recovered by one quadrature (*reconstruction*):*

$$q_1(t) = q_1(0) + \int_0^t \frac{\partial H}{\partial p_1}\big(q'(s), \ell, p'(s)\big)\,\mathrm{d}s. \tag{10.8}$$

Proof. We have $\dot{p}_1 = -\partial H/\partial q_1 = 0$. With $p_1 = \ell$ inserted, the equations $\dot{q}_i = \partial H/\partial p_i$ and $\dot{p}_i = -\partial H/\partial q_i$ for $i \geq 2$ are those of H_ℓ, because setting $p_1 = \ell$ commutes with differentiating in q_i, p_i for $i \geq 2$. They no longer involve q_1. The remaining equation $\dot{q}_1 = \partial H/\partial p_1$ has a right-hand side that is now a known function of t. Integrating gives the reconstruction formula. ∎

Reduction has two steps.

- *Level set.* The motion stays on $\{p_1 = \ell\}$, which has dimension $2n-1$.
- *Quotient.* The symmetry — translation of q_1 — is the flow of the conserved quantity itself: $X_{p_1} = \partial/\partial q_1$. In the language of § 10.2, p_1 is the momentum map of the translations, the lift (10.6) with $\xi_U = e_1$. Its flow moves points *along* the level set, and the reduced variables (q', p') label its orbits. Forgetting q_1 is dividing by the symmetry: dimension $2n-2$.

Example 10.11 (The planar central force)**.** Consider a particle of mass m in the punctured plane with $H = |p|^2/2m + V(r)$, $V \in C^2(0,\infty)$. Polar coordinates (r, φ) are a point transformation, so their lift is canonical by the point-transformation example of Example 8.11,

and Hamilton's equations keep their form, with the same H written in the new variables (Theorem 8.12, with F_2 independent of t, on any sector where φ is single-valued). The new momenta are

$$p_r = p \cdot \frac{\partial q}{\partial r} = \frac{q \cdot p}{r}, \qquad p_\varphi = p \cdot \frac{\partial q}{\partial \varphi} = q_1 p_2 - q_2 p_1 = L_3, \qquad H = \frac{p_r^2}{2m} + \frac{p_\varphi^2}{2mr^2} + V(r).$$

The angle φ is cyclic, so $L_3 = \ell$ is conserved; (r, p_r) obey Hamilton's equations for

$$H_\ell(r, p_r) = \frac{p_r^2}{2m} + V_{\text{eff}}(r), \qquad V_{\text{eff}}(r) = \frac{\ell^2}{2mr^2} + V(r), \tag{10.9}$$

and (10.8) reads $\dot\varphi = \ell/(mr^2)$. Four dimensions become two.

Substitution must be made in the Hamiltonian. The constant ℓ goes into H, where p_φ is one of the coordinates. It may not be put into the Lagrangian $\mathcal{L} = \frac{1}{2}m(\dot r^2 + r^2\dot\varphi^2) - V(r)$ through $\dot\varphi = \ell/(mr^2)$: that gives $\frac{1}{2}m\dot r^2 - \big(V - \ell^2/(2mr^2)\big)$, whose centrifugal term has the wrong sign.

10.5 The central force, reduced

In space the symmetry group is SO(3). Let $\mathcal{M} = (\mathbb{R}^3 \setminus 0) \times \mathbb{R}^3$ and $H = |p|^2/2m + V(|q|)$, $V \in C^2(0,\infty)$. The rotations act by the lifted maps $(q,p) \mapsto (Rq, Rp)$, H is invariant, and by Theorem 10.9 all three components of the momentum map $J = \mathbf{L} = q \times p$ are conserved. Fix a value $\boldsymbol{\ell} \neq 0$ and write $\ell = |\boldsymbol{\ell}|$.

Six dimensions, three conserved components, and a three-dimensional group: why does the reduced system not have $6 - 3 - 3 = 0$ dimensions? A rotation R carries the level set $\{\mathbf{L} = \boldsymbol{\ell}\}$ to $\{\mathbf{L} = R\boldsymbol{\ell}\}$, since $Rq \times Rp = R(q \times p)$. Only the rotations about $\boldsymbol{\ell}$ keep it, and they form a single direction. Thus $6 - 3 - 1 = 2$. The theorem below identifies the two reduced coordinates.

Theorem 10.12 (The central force, reduced)**.** *Let a motion have $\mathbf{L} = \boldsymbol{\ell} \neq 0$, and choose the axes so that $\boldsymbol{\ell} = \ell e_3$. Then the motion lies in the plane $\{q_3 = p_3 = 0\}$, and in the polar coordinates $(r, \varphi, p_r, p_\varphi)$ of that plane $p_\varphi = \ell$ and:*

1. *$(r(t), p_r(t))$ solves Hamilton's equations, with $\omega_\ell = \mathrm{d}r \wedge \mathrm{d}p_r$, for the reduced Hamiltonian H_ℓ of (10.9);*
2. *the azimuth is reconstructed from $\dot\varphi = \ell/(mr^2)$.*

Proof. A rotation of the axes is a lifted point transformation, hence canonical, and it leaves H unchanged, so the axes may be chosen so. Since q and p are orthogonal to $q \times p = \ell e_3$, $q_3 = p_3 = 0$ along the motion (the planarity of Proposition 3.16). At such points $\partial H/\partial q_3 = V'(|q|)\, q_3/|q| = 0$ and $\partial H/\partial p_3 = p_3/m = 0$, so the remaining four of Hamilton's equations are those of the planar Hamiltonian of Example 10.11, with $p_\varphi = L_3 = \ell$. Theorem 10.10 does the rest. ∎

The planar $\omega = \mathrm{d}r \wedge \mathrm{d}p_r + \mathrm{d}\varphi \wedge \mathrm{d}p_\varphi$ records one oriented area for each pair (§ 6.5). On the level set the coordinate p_φ is the constant ℓ, so only the area in the (r, p_r) plane is left, which is $\omega_\ell = \mathrm{d}r \wedge \mathrm{d}p_r$. This forgets exactly the symmetry direction $\partial/\partial\varphi$.

The reduced system is a single degree of freedom, so Lecture 1's method applies: draw V_{eff} and read off the portrait (§ 1.4). The term $\ell^2/2mr^2$ is the centrifugal barrier. A non-degenerate minimum r_0 of V_{eff} is a centre of the reduced portrait: an equilibrium of the

reduced system, and a circular orbit in space. A closed reduced curve is a motion whose distance oscillates between two turning points while $\dot{\varphi} = \ell/mr^2 > 0$ drives it round the centre.

10.6 Kepler: its orbits and its hidden symmetry

Take $V = -k/r$ with $k > 0$. The effective potential $V_{\text{eff}} = \ell^2/2mr^2 - k/r$ has a single critical point, a non-degenerate minimum:

$$r_0 = \frac{\ell^2}{mk}, \qquad V_{\text{eff}}(r_0) = -\frac{mk^2}{2\ell^2}, \qquad V''_{\text{eff}}(r_0) = \frac{m^3k^4}{\ell^6} > 0.$$

The portrait in Figure 10.2 gives the following motions. The level $E = V_{\text{eff}}(r_0)$ is the circular orbit; for $V_{\text{eff}}(r_0) < E < 0$ the reduced orbit is a closed curve between two turning points $r_- < r_+$, periodic by Proposition 1.13; for $E \geq 0$ the particle escapes to $r = \infty$.

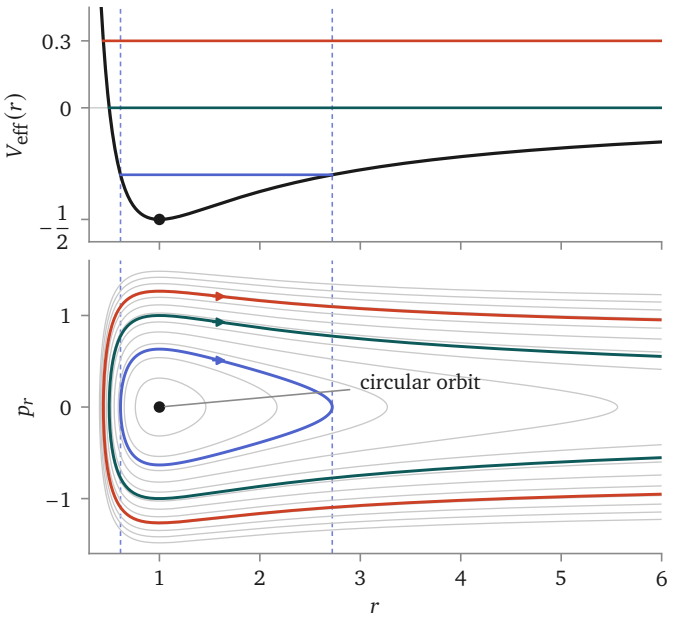


Figure 10.2: Kepler, reduced ($m = k = \ell = 1$). Top: V_{eff}, with the levels $E = -0.3$ (blue; dashed lines carry its turning points $1/(1 \pm \sqrt{0.4})$ down), $E = 0$ and $E = 0.3$. Bottom: the reduced portrait; the dot is the circular orbit.

The reduced motion is periodic, but the orbit in space need not close: in one radial period the azimuth advances by an angle which, for a general V, varies with E and ℓ, and unless it is a rational multiple of 2π the orbit never closes. Kepler's bounded orbits (with $\ell \neq 0$, as throughout) all close. The reason is one more conserved quantity. We write it down and verify it; we do not derive it.

The Runge–Lenz vector

Kepler's *Laplace–Runge–Lenz vector* is

$$\mathbf{A} = p \times \mathbf{L} - mk\,\frac{q}{|q|}. \tag{10.10}$$

Proposition 10.13 (Runge–Lenz). *The vector* $\mathbf{A}$ *is conserved, and* $\mathbf{A} \cdot \mathbf{L} = 0$.

Proof. By Proposition 5.11 it suffices to show that $\mathbf{A}$ is constant along each motion. There $\dot{q} = p/m$, $\dot{p} = -kq/|q|^3$ and $\dot{\mathbf{L}} = 0$. With $r = |q|$,

$$\dot{p} \times \mathbf{L} = -\frac{k}{r^3}\, q \times (q \times p) = \frac{k}{r}\, p - \frac{k\,(q \cdot p)}{r^3}\, q = mk\, \frac{\mathrm{d}}{\mathrm{d}t} \frac{q}{r},$$

so $\dot{\mathbf{A}} = 0$. Next, $p \times \mathbf{L}$ and q are orthogonal to $\mathbf{L} = q \times p$. ∎

Proposition 10.14 (Kepler's orbits). *Let* $V = -k/r$, *and let a motion have* $\mathbf{L} \neq 0$, *with the axes and polar coordinates of Theorem 10.12. Then*

$$r = \frac{\ell^2}{mk + |\mathbf{A}| \cos(\varphi - \varphi_{\mathbf{A}})}, \tag{10.11}$$

where, if $\mathbf{A} \neq 0$, $\varphi_{\mathbf{A}}$ *is the direction of* $\mathbf{A}$ *in the plane of the orbit (if* $\mathbf{A} = 0$ *the cosine term vanishes, and any angle will do). If moreover* $E < 0$, *then* $|\mathbf{A}| < mk$: *the orbit is an ellipse with a focus at* 0 *and eccentricity* $|\mathbf{A}|\,/mk$ *(the circle if* $\mathbf{A} = 0$*).*

Proof. The vector $\mathbf{A}$ is constant and $\mathbf{A} \cdot \mathbf{L} = 0$ (Proposition 10.13), so $\mathbf{A}$ is a fixed vector in the plane of the orbit. By the cyclic property of the triple product,

$$\mathbf{A} \cdot q = q \cdot (p \times \mathbf{L}) - mk\,|q| = \mathbf{L} \cdot (q \times p) - mkr = \ell^2 - mkr.$$

The left side is $|\mathbf{A}|\; r\cos(\varphi - \varphi_{\mathbf{A}})$ (any $\varphi_{\mathbf{A}}$ if $\mathbf{A} = 0$), so $r\big(mk + |\mathbf{A}|\cos(\varphi - \varphi_{\mathbf{A}})\big) = \ell^2 > 0$, which gives (10.11); in particular the bracket is positive at every azimuth the motion reaches.

Let $E < 0$. By the reduced portrait $r(t)$ stays in $[r_-, r_+] \subset (0, \infty)$ (with $r_\pm = r_0$ on the circle), so $\dot{\varphi} = \ell/mr^2 \geq \ell/mr_+^2 > 0$ and φ takes every value. The bracket is therefore positive for every φ, which forces $|\mathbf{A}| < mk$, and (10.11) is the polar equation of the ellipse stated. ∎

So a bounded Kepler orbit is an ellipse, traced once in each turn of φ: it closes. Unless it is the circle $\mathbf{A} = 0$, $\mathbf{A}$ points to the pericentre, where r is least (Figure 10.3).

The Runge–Lenz vector does not generate a point symmetry

Can $\mathbf{A}$ be the charge of a symmetry of positions? Its momentum dependence rules this out.

Proposition 10.15 (Not a point symmetry). *No component of* $\mathbf{A}$ *is, up to a constant, a Noether charge* $p \cdot \xi(q,t) - G(q,t)$ *of Theorem 3.14 or a momentum map* $p \cdot \xi_U(q)$ *of a lifted action (Theorem 10.6).*

Proof. Those are affine in p, while $A_x = q_x(p_y^2 + p_z^2) - p_x(q_y p_y + q_z p_z) - mkq_x/|q|$ is quadratic in p, and so, by the same formula with the axes permuted, are A_y and A_z. ∎

Still, $\{A_a, H\} = 0$, so by Noether reciprocity (Theorem 6.9) H *is invariant under the flow of each* A_a. That flow is a symmetry of H which is not induced by a point transformation of configuration space.

Summary.

- *Symmetry and momentum:* a symplectic action of a matrix group, its generators (Lemma 10.4) and its momentum map (Definition 10.5). A lifted action has $J_\xi = p \cdot \xi_U(q)$ (Theorem 10.6), and rotations have $J = \mathbf{L} = q \times p$ (Example 10.8). An invariant H conserves every component of J (Theorem 10.9).

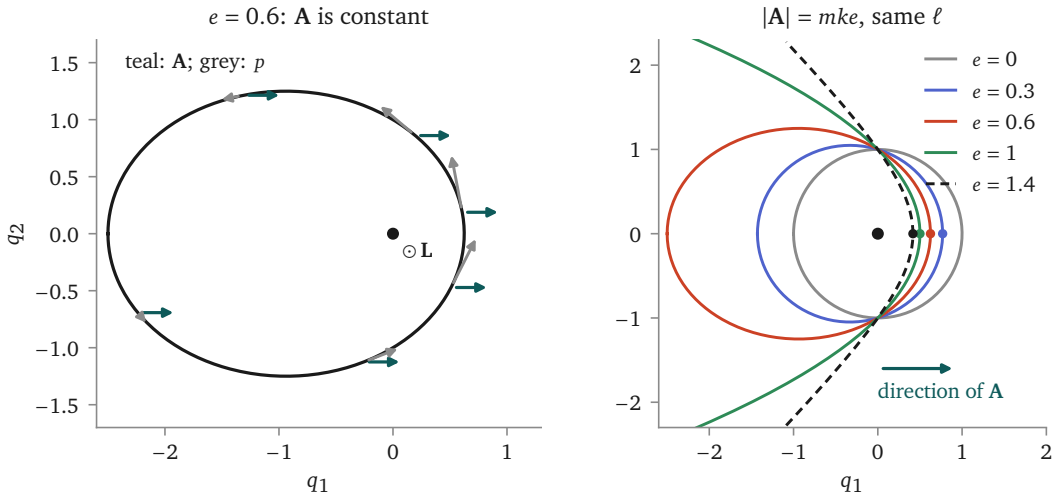


Figure 10.3: Kepler orbits, $m = k = |\mathbf{L}| = 1$, $\mathbf{L}$ out of the page. Left: along the ellipse p turns and $\mathbf{A}$ (drawn at six points) does not. Right: orbits with the same $|\mathbf{L}|$: the circle $e = 0$, where $\mathbf{A} = 0$, and orbits whose $\mathbf{A}$ has one common direction, pointing to the pericentre (dots). Here $e = |\mathbf{A}|/mk$, and orbits with $e \geq 1$ have $E \geq 0$ and are unbounded.

- *Reduction:* a cyclic variable reduces by one degree of freedom (Theorem 10.10), and the central force with $\mathbf{L} \neq 0$ reduces to a motion in V_{eff} (Theorem 10.12).
- *Kepler:* the vector $\mathbf{A}$ closes every bounded orbit with $\mathbf{L} \neq 0$ (Proposition 10.14), though no point symmetry produces it (Proposition 10.15).
- *Next:* the rigid body has the rotations as its symmetry too. Dividing them out completely leaves a bracket that is degenerate, and Lecture 11 studies such brackets.

References

Marsden and Ratiu, *Introduction to Mechanics and Symmetry*, §9.3 (actions, p. 309), §11.2 (the definition of a momentum map, p. 367) and Theorem 12.1.4 (lifted actions, p. 386); their conventions are ours. Arnold, *Mathematical Methods of Classical Mechanics*, Appendix 5A (the lift formula, p. 371), §8 (the effective potential, p. 34) and §15C (cyclic coordinates, p. 67); his ω is $\mathrm{d}p \wedge \mathrm{d}q$, which nothing here uses. Tong, *Lectures on Classical Dynamics*, p. 96, for the orbit from the Runge–Lenz vector.